Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

[AIME-2021-II-Q11] | Part C (Q11-Q15)

Answer: 258

1. List all valid four-integer runs allowed by the announcement. A student can initially identify the run exactly when their number occurs in only one candidate run. Thus the unanimous failure removes every run containing such a unique number.

2. Enumerating the starts 10 through 96 leaves exactly the runs $[34, 37], [47, 50], [76, 79], [89, 92]$. Equivalently these have the consecutive 6- and 7-multiples in their two middle positions.

3. The four surviving runs are disjoint, so every number now identifies its run, meeting the second condition. Their largest values sum to $37 + 50 + 79 + 92 = 258$.

Source / 来源